

# Random Variable Generation and Performance Evaluation

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## 1 Introduction

In this laboratory assignment, we designed and implemented functions to generate several classes of random variables (RVs) commonly used in stochastic modeling. In particular, we focused on:

- **Hyper-exponential random variables**, implemented using a *composition* method.
- **Erlang- $k$  random variables**, implemented via *convolution* of exponential random variables.
- **Pareto random variables**, generated using the *inverse transform* method.

These RVs were subsequently used for testing and performance evaluation in queueing systems and a simulated car-sharing environment.

## 2 Random Variable Implementation

### 2.1 Hyper-exponential Distribution

The hyper-exponential distribution is a mixture of  $n$  exponential distributions and is used to model highly variable processes. Its PDF is

$$f_X(x) = \sum_{i=1}^n p_i \lambda_i e^{-\lambda_i x}, \quad x \geq 0, \quad \sum_{i=1}^n p_i = 1,$$

where  $p_i$  is the probability of selecting branch  $i$  and  $\lambda_i = 1/\mu_i$ .

#### Generation: Composition Method

1. Select a branch  $i$  according to  $p$ .
2. Generate  $X \sim \text{Exponential}(\lambda_i)$  for the selected branch.

Equivalently,

$$X = \sum_{i=1}^n \mathbf{1}_{\{I=i\}} Y_i, \quad Y_i \sim \text{Exponential}(\lambda_i),$$

where  $\mathbf{1}_{\{I=i\}}$  indicates the chosen branch.

### 2.2 Erlang- $k$ Distribution

An Erlang- $k$  random variable is defined as the sum of  $k$  independent exponential random variables with common rate

$$\lambda = \frac{k}{\mu},$$

where  $\mu$  is the desired mean of the Erlang distribution. Thus, if  $Y_1, \dots, Y_k$  are i.i.d. exponential variables with parameter  $\lambda$ , the Erlang variable is generated through the convolution method.

### Generation: Convolution Method

1. Generate  $k$  i.i.d. exponential variables  $Y_i \sim \text{Exponential}(\lambda)$ .
2. Sum them:  $X = \sum_{i=1}^k Y_i$ .

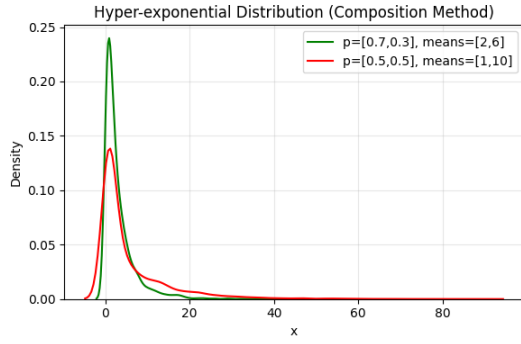
## 2.3 Pareto Distribution

Using the inverse transform method, a uniform variable  $U \sim \text{Uniform}(0, 1)$  is converted to a Pareto sample. The PDF is

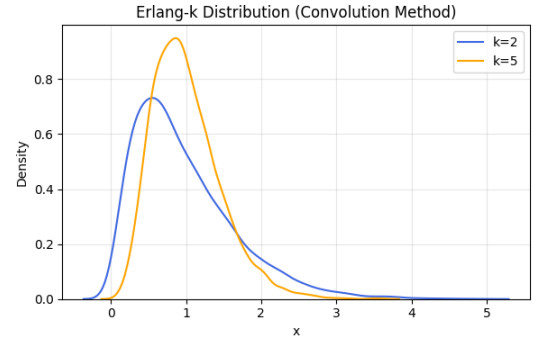
$$f_X(x) = \frac{\alpha x_m^\alpha}{x^{\alpha+1}}$$

### Generation: Inverse Transform

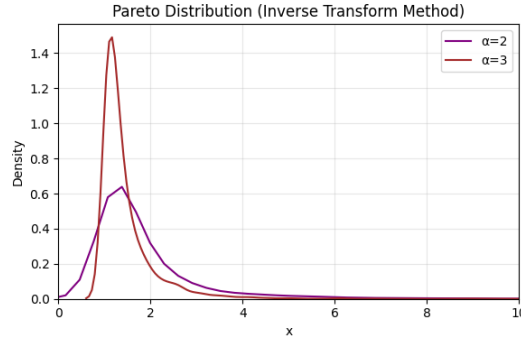
1. Sample  $U \sim \text{Uniform}(0, 1)$ .
2. Transform:  $X = F_X^{-1}(U) = x_m(1 - U)^{-1/\alpha}$ .



(a) Hyper-exponential



(b) Erlang- $k$



(c) Pareto

Figure 1: Comparison of the three random variable distributions.

Samples were generated from Erlang, hyper-exponential and Pareto distributions using a random seed for reproducibility, with  $n = 5000$  samples each.

The hyper-exponential distribution shows high variability and occasional long intervals. The green line, corresponding to the first parameter set, is more concentrated with shorter intervals, while the red line, corresponding to the second parameter set, is more spread out with longer intervals, reflecting greater burstiness.

For the Erlang distribution, increasing  $k$  from 2 to 5 makes the distribution more concentrated around the mean, reducing variability.

The Pareto distribution, with shape parameters  $\alpha = 2$  and  $\alpha = 3$ , exhibits heavy tails: smaller  $\alpha$  produces more extreme large values, while larger  $\alpha$  concentrates samples closer to the minimum  $x_m$ .

### 3 Testing and Validation

Two types of statistical tests were performed to verify the correctness and independence of the generated samples.

#### 3.1 Test 1: Distribution Accuracy

To check that the generated random variables follow the intended distribution, we applied the Kolmogorov-Smirnov (KS) test. This test compares the empirical cumulative distribution function (CDF) of the samples to the theoretical CDF. For Erlang- $k$  samples, the KS test returned a statistic of 0.0126 and a p-value of 0.4, indicating that the null hypothesis (samples follow the theoretical distribution) could not be rejected. Figure 2a shows the histogram of the empirical samples against the theoretical PDF.

#### 3.2 Test 2: Independence of Samples

Independence of samples was assessed using the autocorrelation function (ACF). Figure 2b shows the lagged autocorrelations of Erlang- $k$  samples up to lag 50. The autocorrelation at lag 0 is always 1 by definition, while the autocorrelations at all other lags are close to zero. This confirms that the generated samples are statistically independent, as expected from the convolution of independent exponential random variables.

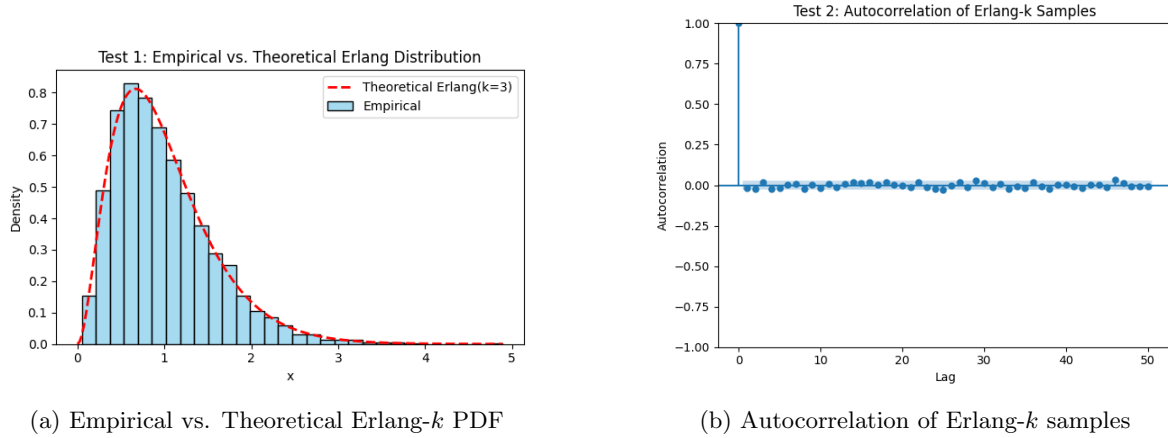


Figure 2: Statistical validation of Erlang- $k$  random variables.

## 4 Performance Evaluation in Queueing Systems

### 4.1 Performance Evaluation of a G/G/1 Queue

To evaluate the impact of different inter-arrival time distributions on queueing performance, we simulated a single-server G/G/1 queue. The simulation proceeds as follows:

1. The system starts empty at time  $t = 0$ .
2. Arrivals are generated sequentially using a specified inter-arrival time distribution (Exponential, Erlang- $k$ , Hyper-exponential, or Pareto), and service times are drawn independently from an exponential distribution with mean 0.8.
3. If the server is idle upon arrival, the incoming customer is immediately served; otherwise, they join a queue.
4. Departures occur after a customer completes service. Waiting times are recorded for all customers who experience a queue.
5. The simulation continues until a total of  $T = 30,000$  time units have elapsed, and the average waiting time is computed.

## 4.2 Performance Evaluation: Single-Server Queue

The average waiting times observed for each inter-arrival distribution were:

Table 1: Queue Performance Metrics for Different Inter-Arrival Distributions

| Distribution      | Mean Waiting Time | Variance | 95th Percentile | Max Waiting Time |
|-------------------|-------------------|----------|-----------------|------------------|
| Exponential       | 3.52              | 11.28    | 10.54           | 19.25            |
| Erlang            | 2.43              | 4.96     | 6.91            | 14.08            |
| Hyper-exponential | 3.87              | 12.16    | 10.92           | 23.38            |
| Pareto            | 3.44              | 10.35    | 10.30           | 17.79            |

The results show how the variability of inter-arrival times impacts queue performance. Erlang arrivals, being more regular and less variable, produced the shortest mean waiting time (2.43 units) and lower 95th percentile (6.91), indicating fewer extreme delays. Exponential arrivals, with moderate variability, resulted in longer average waiting times (3.52) and higher variance (11.28). Hyper-exponential arrivals, characterized by high variance, led to the largest maximum waiting time (23.38) and a high mean (3.87), reflecting the destabilizing effect of irregular arrivals. Pareto arrivals, although heavy-tailed, had a mean waiting time (3.44) slightly lower than hyper-exponential; the occasional long inter-arrival gaps allowed the system to clear accumulated jobs, reducing congestion in some cases. Overall, these findings are consistent with queueing theory: lower variability in arrivals leads to reduced congestion and shorter waiting times, while higher variability increases both mean and extreme waiting times.

## 4.3 Performance Evaluation of the Car-Sharing System

To study the effect of inter-arrival time distributions on the car-sharing system, we simulated a fleet of vehicles serving multiple city hotspots and recorded the system performance in terms of service rate, dropped requests, and average user waiting time.

Table 2 summarizes the simulation results for different inter-arrival time distributions.

Table 2: Car-Sharing System Performance Metrics under Different Arrival Distributions

| Distribution      | Service Rate | Dropped Requests | Average Waiting Time (min) |
|-------------------|--------------|------------------|----------------------------|
| Exponential       | 67.16%       | 5260             | 13.08                      |
| Erlang            | 68.14%       | 5036             | 13.03                      |
| Hyper-exponential | 69.20%       | 3676             | 12.74                      |
| Pareto            | 68.73%       | 5000             | 12.83                      |

The results indicate clear performance differences across the four arrival distributions. The hyper-exponential distribution produces the highest service rate and the lowest number of dropped requests, reflecting how its high variability occasionally generates long inter-arrival gaps that allow the system to recover from congestion and serve more users. The Pareto distribution, although also heavy-tailed, performs slightly worse than the hyper-exponential: it still reduces waiting times and congestion compared to exponential arrivals, but its intermittently large bursts lead to more dropped requests.

Erlang arrivals, which are more regular and less variable, generate moderate congestion levels, resulting in a slightly lower service rate but still better performance than the exponential case. Exponential arrivals, with intermediate variability, lead to the longest waiting times and highest number of dropped requests among the four, making it the least efficient in this scenario.

Overall, the results show that variability in inter-arrival times can sometimes improve system performance by creating recovery periods in the queue, though extremely heavy-tailed distributions may not always be as beneficial depending on their specific parameters.

## 5 Conclusion

In this lab, hyper-exponential, Erlang- $k$ , and Pareto random variables were implemented using composition, convolution, and inverse transform methods, respectively. Statistical tests, including the

Kolmogorov-Smirnov test and autocorrelation analysis, confirmed the accuracy and independence of the generated samples. Performance evaluation in both a single-server queue and a car-sharing system showed that the variability of inter-arrival times strongly influences service metrics. Erlang arrivals, being more regular, slightly improved waiting times, while hyper-exponential and Pareto arrivals, which exhibit higher variability, sometimes increased the service rate by allowing temporary gaps that reduce congestion. These results highlight that understanding the stochastic properties of arrival processes is crucial for designing efficient service systems, as variability can both hinder and improve performance depending on the context.

# Simulation of Airport Passenger Flow: Multi-Server Queueing Model

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## Abstract

This report presents a detailed discrete-event simulation of passenger flow in a medium-sized airport, capturing passenger arrival dynamics, service nodes, and commercial facilities. Passenger arrivals are modeled using a mixture of stochastic distributions reflecting realistic early arrivals and peak hours. Multi-server adaptive queues model critical service nodes such as bag drop and security, while commercial facilities are visited probabilistically without queuing. Monte Carlo simulations are performed to estimate confidence intervals for node metrics. Results include node utilization, waiting and service times, arrival profiles, and heatmaps of passenger distribution across the airport.

## 1 Introduction

Efficient airport operations require understanding passenger flows to reduce congestion and improve passenger experience. Key service points such as bag drop counters and security checkpoints often experience variable loads, while commercial facilities attract passengers probabilistically. This study simulates passenger flow in a medium-sized airport, capturing stochastic arrivals, variable service times, and probabilistic visits to shops and restaurants.

## 2 Theoretical Model

The airport is modeled as a directed network of nodes representing critical service points and commercial facilities:

- **Entrance:** Starting point for all passengers.
- **Bag Drop:** Multi-server (M/M/c) queue handling checked baggage.
- **Security Check:** Multi-server (M/M/c) queue for all passengers.
- **Shops and Restaurants:** Landside (before security) and airside (after security) facilities.
- **Gates:** Final destination for boarding.

Node positions are mapped on a  $7 \times 7$  grid to facilitate visualization, and travel times between nodes are explicitly defined. The airport is considered closed between 00:00 and 05:00, during which no passengers are present.

Travel times between key nodes are:

Entrance  $\rightarrow$  Bagdrop = 2 min,                      Bagdrop  $\rightarrow$  Security = 3 min,  
Bagdrop  $\rightarrow$  Shops Landside = 1 min,      Bagdrop  $\rightarrow$  Restaurants Landside = 1 min,  
Security  $\rightarrow$  Shops Airside = 1 min,              Security  $\rightarrow$  Restaurants Airside = 1 min,  
Airside  $\rightarrow$  Gate = 4 min.

The airport is considered closed between 00:00 and 05:00, during which no passenger arrivals are scheduled.

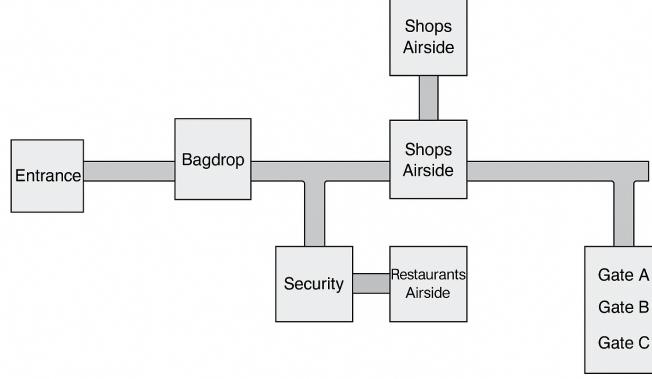


Figure 1: Schematic of airport nodes: entrance, bag drop, security, shops, restaurants, and gates.

## 2.1 Flight and Passenger Generation

The simulation is performed for a single day ( $\text{DAYS} = 1$ ) with 45 scheduled flights ( $\text{FLIGHTS\_PER\_DAY} = 45$ ). Each flight carries an average of 160 passengers ( $\text{AVG\_PAX\_PER\_FLIGHT} = 160$ ), with the exact number sampled from a binomial distribution:

$$N_{\text{flight}} \sim \text{Binomial}(n = 160, p = 0.85). \quad (1)$$

Passengers carry checked baggage with probability  $\text{BAG\_PROB} = 0.62$ , determining whether they visit Bagdrop before Security or proceed directly to Security.

Passenger arrivals are sampled from a mixture distribution to reflect realistic early arrivals and peak hours:

$$T_{\text{arrival}} \sim \begin{cases} 70\% & U(0, 24 \cdot 60) & \text{(uniform arrivals throughout the day)} \\ 15\% & \mathcal{N}(480, 40^2) & \text{(morning peak around 08:00)} \\ 15\% & \mathcal{N}(1140, 40^2) & \text{(evening peak around 19:00)} \end{cases} \quad (2)$$

where times are in minutes from midnight and arrivals are shifted to avoid the closed interval (00:00–05:00).

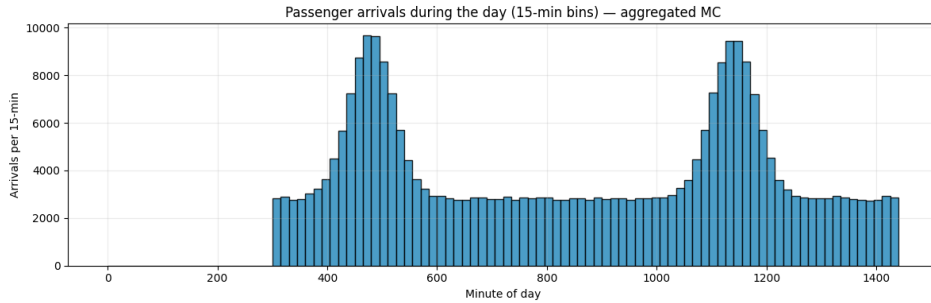


Figure 2: Passenger arrival profile throughout the day (histogram aggregated over Monte Carlo runs).

## 2.2 Passenger Behavior and Service

Passenger behavior through the network is probabilistic. After Bagdrop and Security, passengers may visit commercial facilities with the following probabilities:

$$\begin{aligned}
P_{\text{shop\_land}} &= 0.25, & P_{\text{rest\_land}} &= 0.18, \\
P_{\text{shop\_air}} &= 0.30, & P_{\text{rest\_air}} &= 0.25.
\end{aligned}$$

Service times are sampled from uniform distributions:

$$\begin{aligned}
S_{\text{bagdrop}}, S_{\text{security}} &\sim U(1, 5) \text{ min}, \\
S_{\text{shop\_land}}, S_{\text{shop\_air}} &\sim U(6, 20) \text{ min}, \\
S_{\text{rest\_land}} &\sim U(20, 50) \text{ min}, \\
S_{\text{rest\_air}} &\sim U(15, 40) \text{ min}.
\end{aligned}$$

Critical service nodes (Bagdrop and Security) are modeled as multi-server queues (M/M/c) where passengers are served in FIFO order. Shops and restaurants do not maintain queues; only the time spent by passengers in each facility is recorded. After all activities, 95% of passengers proceed to their assigned gate, while 5% may not reach a gate.

Each passenger maintains a complete history of node visits, including waiting times, service times, and total durations. This allows capturing stochastic arrivals, queue formation, service dynamics, and probabilistic passenger behavior.

### 2.3 Monte Carlo Simulation and Confidence Intervals

To account for stochastic variability, the simulation is repeated 50 times ( $N_{\text{SIM}} = 50$ ). For each node, the following metrics are recorded:

- Unique visitors per node (excluding gates in summary table)
- Mean time spent per node
- Per-minute occupancy (used for time-series plots)

After all Monte Carlo runs, 95% confidence intervals are computed across the  $N_{\text{SIM}}$  runs for both metrics:

$$\text{CI}_{95\%} = \bar{x}_i \pm 1.96 \frac{s_i}{\sqrt{N_{\text{SIM}}}}, \quad (3)$$

where  $\bar{x}_i$  is the average value of the metric for node  $i$  across all runs,  $s_i$  is the standard deviation across runs, and  $N_{\text{SIM}}$  is the total number of Monte Carlo simulations.

To ensure reproducibility, each Monte Carlo simulation is initialized with a **random seed**. The base seed guarantees that results can be replicated, while for each run the seed is slightly changed so that stochastic elements—such as passenger arrivals, service times, and routing decisions—vary between runs.

This procedure provides statistically robust estimates of node-level passenger flow and service times, capturing the variability due to stochastic arrivals, service processes and probabilistic passenger routing.

## 3 Implementation and Event Dynamics

The simulation is implemented using a *Future Event Set (FES)*. Nodes correspond to functional areas and edges represent travel paths with explicit travel times.

- **Nodes:** Each node is implemented as a class that manages a queue of waiting passengers and the servers providing service.
  - **Bagdrop:** Multi-server queue (M/M/c) with a fixed number of 15 servers. Passengers join the queue and are served in the order of arrival (FIFO discipline).
  - **Security:** Multi-server queue (M/M/c) with a fixed number of 20 servers, also using FIFO service.
  - **Shops and Restaurants:** No queues are modeled; only the time passengers spend in these facilities is recorded, sampled from stochastic service-time distributions.



- **Passengers:** Represented by the `Passenger` class, which tracks each visit to a node. For each node, the passenger records arrival time, start and end of service, waiting time, and total time spent. The full visit history is maintained, allowing post-simulation analyses such as mean times, occupancy, and flow patterns.
- **Events:** The FES drives the simulation by scheduling and processing events including:
  - Arrival at a node
  - End of service at a node
  - Movement between nodes

When a passenger arrives at a node, they enter the queue. If a server is available, service begins immediately; otherwise, the passenger waits. Completing service frees a server and may trigger the next node visit according to the passenger’s probabilistic path. Travel times along edges are explicitly modeled, ensuring realistic temporal dynamics.

This structure provides a modular and efficient implementation capable of realistically capturing queue formation, service dynamics, passenger movement, and the spatial distribution of visitors throughout the terminal.

## 4 Simulation Results

### 4.1 Node Metrics with Confidence Intervals

Table 3 shows the mean visitors and mean time per node across 50 simulation runs with 95% confidence intervals.

|           | mean_visitors | visitors_CI        | mean_time_min      | time_CI        |
|-----------|---------------|--------------------|--------------------|----------------|
| bagdrop   | 3790.56       | [3776.35, 3804.77] | 6.829306115391128  | [6.70, 6.96]   |
| security  | 6116.18       | [6107.97, 6124.39] | 13.578216765483376 | [13.35, 13.81] |
| shop_land | 950.76        | [943.57, 957.95]   | 13.016907390862961 | [12.98, 13.05] |
| rest_land | 684.6         | [678.11, 691.09]   | 35.0170695216373   | [34.92, 35.11] |
| shop_air  | 1835.88       | [1825.12, 1846.64] | 13.005731351455282 | [12.98, 13.03] |
| rest_air  | 1528.4        | [1518.77, 1538.03] | 27.498992488863056 | [27.45, 27.55] |

Figure 3: Node metrics with 95% confidence intervals from Monte Carlo simulations.

**Discussion:** - The bag drop and security nodes handle the majority of passengers, as expected. Average times are moderate, reflecting efficient service with adaptive servers. - The commercial facilities show average times consistent with expectations: shopping around 13 minutes, landside restaurants around 35 minutes, airside restaurants around 27 minutes. This aligns with realistic passenger behavior and the uniform service time assumptions. - Confidence intervals are narrow, showing the simulation is stable across multiple runs.

### 4.2 Passenger Affluence per Node

Figure 4 presents the number of passengers present in each node over time, aggregated in 5-minute bins. This shows peak congestion periods and low activity overnight. Note that the airport is considered closed between 00:00 and 5:00, so no passengers are present in this interval.

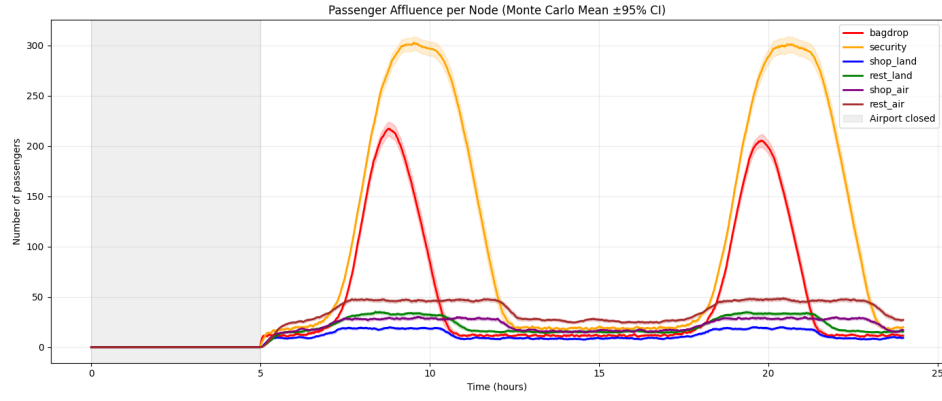


Figure 4: Passenger affluence per node over time.

#### Discussion:

- Bagdrop and Security exhibit clear peaks in passenger numbers that correspond to peak hours, reflecting the periods of highest demand rather than the entire day.
- Commercial facilities have smaller, broader peaks, reflecting optional visits.
- Overnight, no passengers are present due to airport closure, consistent with expectations.

### 4.3 Heatmap of Unique Visitors per Node

Figure 5 presents the spatial distribution of passengers across the airport. Bag drop and security dominate visitation, while commercial facilities are visited by fewer passengers, consistent with probabilistic behavior.

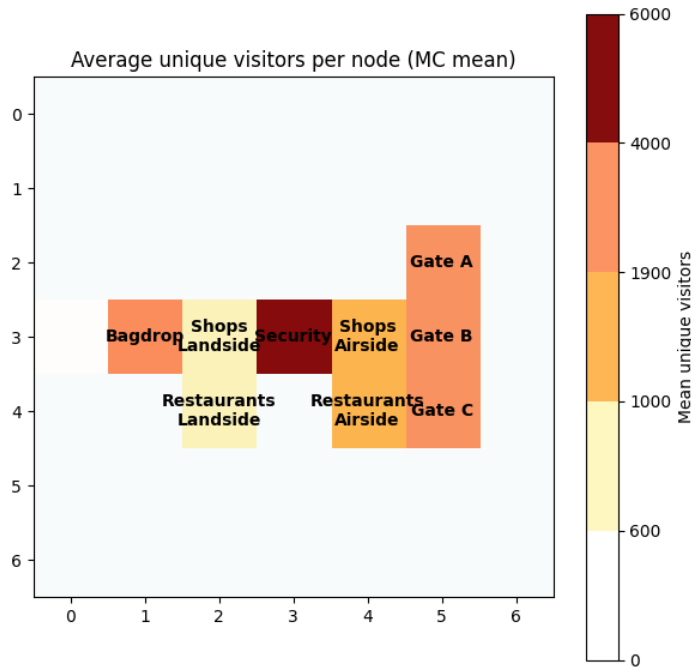


Figure 5: Heatmap of unique visitors per node. Red intensity indicates higher counts.

## 5 Analysis

Bag drop and security experience moderate to high congestion, mitigated by adaptive servers. Commercial facilities have service times reflecting realistic visit durations. The Monte Carlo simulation provides confidence intervals, quantifying variability in node metrics. Passenger affluence plots reveal temporal dynamics, including peak and low periods, confirming that stochastic arrivals and probabilistic behavior are accurately captured.

## 6 Conclusion

The simulation captures essential aspects of airport passenger flow, including stochastic arrivals, multi-server queues and probabilistic visits to shops and restaurants. However, several extensions could improve realism:

- **Dynamic flight schedules:** Incorporating variable departure times and delays to reflect real operational variability.
- **Passenger heterogeneity:** Modeling differences in walking speeds, service preferences, priority status, or the presence of accompanying passengers at the entrance.
- **Gate arrivals:** Explicitly simulating passenger arrivals at gates to capture boarding dynamics and gate congestion.
- **Detailed terminal modeling:** Creating multiple terminals or sub-terminals, as in large airports like Malpensa, to reflect realistic passenger routing and facility usage.
- **Time-dependent service rates:** Adjusting processing speeds at Bagdrop and Security according to staffing levels or peak-period efficiency.
- **Behavioral factors:** Including passenger choices, such as deciding to skip commercial areas when queues are long or forming groups.

These enhancements would allow the model to more accurately reflect real airport operations and support more informed planning and resource allocation decisions.

# Markovian Simulation of a Predator–Prey Ecosystem: Monte Carlo Analysis and Population Dynamics

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## Abstract

This report presents a detailed description and analysis of a Markovian, event-driven simulator modeling the population dynamics of two animal species: herbivores and carnivores. The model explicitly tracks the states of male and female individuals, accounts for pregnancy, and includes key interactions such as reproduction, predation, intra-species fights and resource competition. We describe the theoretical model, the implementation and the results of multiple simulation tests with varying parameters.

## 1 Introduction

The goal of this simulation is to study the temporal evolution of populations of herbivores (H) and carnivores (C) under stochastic interactions. The system is *Markovian*, meaning that the probability distribution of future states depends only on the current state and not on the past history.

Formally, the system state at time  $t$  is defined as:

$$S(t) = \{a_1(t), a_2(t), \dots, a_{N(t)}(t)\},$$

where  $a_i(t)$  represents a living animal and contains the following attributes:

$$a_i(t) = (\text{species}, \text{gender}, \text{pregnant})$$

In our model, we track explicitly the following state variables:

$$\mathbf{S}(t) = (H, H_m, H_f, \text{preg}H_f, C, C_m, C_f, \text{preg}C_f),$$

where:

$$\begin{aligned} H(t) &= \text{total number of herbivores} \\ H_m(t) &= \text{number of male herbivores} \\ H_f(t) &= \text{number of female herbivores} \\ \text{preg}H_f(t) &= \text{number of pregnant female herbivores} \\ C(t) &= \text{total number of carnivores} \\ C_m(t) &= \text{number of male carnivores} \\ C_f(t) &= \text{number of female carnivores} \\ \text{preg}C_f(t) &= \text{number of pregnant female carnivores} \end{aligned}$$

These variables allow us to model the system dynamics while capturing gender-specific reproductive behavior, inter-species conflicts (predation) and the finite average lifespan of animals.

## 2 Model Dynamics

The model describes the evolution of an ecosystem composed of herbivores and carnivores. Population dynamics emerge from stochastic interactions among individuals, each governed by events scheduled

using exponential distributions to preserve the memoryless property typical of continuous-time Markov processes. The main mechanisms include reproduction, birth, natural death, predation, intraspecific fights and competition-induced mortality.

Population evolution emerges from a collection of stochastic events:

$$\mathcal{E} = \{\text{death, meeting, birth}\},$$

each extracted in chronological order. This ensures a fully memoryless, Markovian evolution.

## 2.1 Natural Death

At birth, each individual draws a lifespan from an exponential distribution:

$$T_{\text{death}} \sim \text{Exp}\left(\frac{1}{\text{scale\_life}}\right), \quad \text{scale\_life} = 8000.$$

A corresponding death event is inserted into the FES.

Natural death ensures that all individuals eventually die even in the absence of ecological pressures such as predation or competition.

## 2.2 Meetings

Each animal schedules its own meeting events with exponential waiting time:

$$T_{\text{meet}} \sim \text{Exp}\left(\frac{1}{\text{Mean\_Meeting\_Time}}\right), \quad \text{Mean\_Meeting\_Time} = 60.$$

When a meeting occurs, the focal animal selects a random partner. Depending on their species and gender, different interactions may occur:

### 2.2.1 Reproduction

When two individuals of the same species and of opposite sex meet, reproduction may occur. If the female is not already pregnant, she becomes pregnant with probability  $p_{\text{reprod}}$ .

Pregnant females cannot reproduce again until the gestation period is completed. Gestation lasts a fixed number of time units:

$$t_{\text{gestation\_end}} = t + G, \quad G = 20.$$

At the gestation end, a birth event is triggered.

## 2.3 Birth

Birth events produce a uniformly distributed litter:

$$L \sim U(1, 6),$$

with newborn gender assigned uniformly. Each newborn immediately receives a scheduled natural death time. Births are strongly affected by competition.

## 2.4 Intraspecific Fights

When two individuals of the same species and the same sex meet, a conflict may occur with probability  $p_{\text{fight}}$ .

The loser is selected uniformly at random and removed from the population. The corresponding counters are updated:

$$\Delta H = -1 \text{ or } \Delta C = -1.$$

This mechanism represents aggression over territorial, reproductive or resource disputes between same-species, same-sex individuals.

## 2.5 Predation

When a carnivore meets a herbivore, the carnivore may successfully kill the herbivore with probability  $p_{\text{pred}}$ :

$$\Delta H = -1.$$

Predation events immediately remove the herbivore from the population. Predation plays a central ecological role: it prevents uncontrolled herbivore growth, provides the primary energy source for carnivores, and strongly influences the long-term balance between the two species. Higher predation probabilities tend to suppress herbivores and support larger carnivore populations, whereas low predation rates may allow herbivores to dominate and eventually trigger competition among them.

## 2.6 Resource Competition

Competition emerges from limited food resources and affects both herbivores and carnivores. Unlike predation or fights, which are triggered by pairwise interactions, competition is a population-level mechanism activated when species abundance crosses ecological thresholds. This competition reduces both fertility and expected lifespan. We apply:

$$\text{remaining lifespan} \leftarrow (1 - 0.30) \cdot \text{remaining lifespan},$$

$$p_{\text{reprod}}^{(H)} \leftarrow 0.6 p_{\text{reprod}}.$$

This models the effect of limited resources in a gradual way, slowing the growth of the herbivore population and increasing the likelihood of natural death during overcrowding. Resource limitation is sufficient to prevent unbounded herbivore growth.

### 2.6.1 Carnivore Competition

Carnivores compete for prey when herbivores become scarce. Competition is triggered when the herbivore-to-carnivore ratio falls below a threshold:

$$\frac{H}{C} < \text{carn\_th},$$

where **carn\_th** is varied across different test scenarios.

When this condition holds, carnivores experience reduced reproductive success and shortened expected lifespan, reflecting limited access to prey.

When competition is active:

- Reduced Fertility: The probability of conception is reduced to 60% of its base value:

$$p_{\text{reprod}}^{(H)} \leftarrow 0.6 \cdot p_{\text{reprod}}.$$

Additionally, the probability of a successful birth event is halved ( $p_{\text{birth}} \leftarrow 0.5 \cdot p_{\text{birth}}$ ).

- Reduced Lifespan: their remaining lifetime is proportionally reduced, increasing the chance of natural death while prey remains scarce.

$$\text{remaining lifespan} \leftarrow (1 - 0.50) \cdot \text{remaining lifespan},$$

which models starvation and reproductive collapse.

This models starvation and intraspecific competition in a gradual way, without imposing instantaneous mortality and captures how insufficient prey limits carnivore survival.

### 3 Metrics

The simulator tracks both final counts and cumulative metrics. Final counts are:

$$\text{Final counts} = \{H, H_m, H_f, \text{preg}H_f, C, C_m, C_f, \text{preg}C_f\}$$

Cumulative metrics include:

- **meetings**: total number of encounters
- **births**: total number of offspring
- **deaths\_total**: sum of all deaths
- **deaths\_natural**: deaths from natural lifespan
- **deaths\_predation**: herbivores killed by carnivores
- **deaths\_fight**: deaths from same-sex fights
- **deaths\_competition**: deaths due to competition
- **fights**: total fights
- **predations**: total predation events
- **realized\_natural\_H**, **realized\_natural\_C**: actual herbivore and carnivore deaths from natural causes

### 4 Implementation Details

#### 4.1 Data Structures

The environment is represented by the class **Environment**, which stores and updates all components needed to track the state of the ecosystem. In particular, it maintains separate lists for herbivores and carnivores, for males and females, and for pregnant females. This structure enables efficient access to species-specific subpopulations, which is essential for handling events such as predation, reproduction, and intraspecific fights.

Each individual is represented by an **Animal** object containing:

- a unique ID,
- the species (herbivore or carnivore),
- gender,
- birth time and scheduled death time,
- pregnancy status (only for females).

These objects allow the simulator to update the state vector

$$\mathbf{S}(t) = (H, H_m, H_f, \text{preg}H_f, C, C_m, C_f, \text{preg}C_f),$$

in a transparent way whenever an event modifies the population.

#### 4.2 Future Event Set (FES)

The temporal evolution of the system is governed by a Future Event Set (FES), a chronologically ordered structure that stores all scheduled events. At any simulation step, the next event in time is extracted and executed, potentially modifying both the population and the FES itself. The main event types are:

- **Death events.** Each animal receives a death time sampled from an exponential distribution with mean equal to its species' lifespan. When a death event occurs, the corresponding state variable is decreased. For example, for a male herbivore:

$$H \leftarrow H - 1, \quad H_m \leftarrow H_m - 1.$$

- **Meeting events.** Two randomly selected animals may interact, leading to different outcomes:

- \* *Predation (carnivore–herbivore):*

$$\Delta H = -1, \quad \Delta C = 0,$$

with the corresponding decrement in  $H_m$  or  $H_f$ .

- \* *Reproduction (male–female of the same species):* With probability  $p_{\text{reprod}}$ , a female becomes pregnant:

$$\text{pregH}_f \leftarrow \text{pregH}_f + 1 \quad \text{or} \quad \text{pregC}_f \leftarrow \text{pregC}_f + 1.$$

- \* *Fight (same species, same sex):* With probability  $p_{\text{fight}}$ , one individual dies:

$$H \leftarrow H - 1 \quad \text{or} \quad C \leftarrow C - 1.$$

- **Birth events.** When a pregnant female reaches the gestation end time, she gives birth to  $k$  newborns (sampled uniformly between 1 and 6). A birth event updates:

$$H \leftarrow H + k, \quad H_f \leftarrow H_f - 1 + (\text{female newborns}), \quad \text{pregH}_f \leftarrow \text{pregH}_f - 1,$$

or analogously for carnivores. New individuals immediately receive scheduled death events, which are inserted into the FES.

### 4.3 State Transitions and Event-Driven Dynamics

Each event produces a local update of the state vector  $\mathbf{S}(t)$  via increments and decrements of the form:

$$\mathbf{S}(t^+) = \mathbf{S}(t^-) + \Delta \mathbf{S}_{\text{event}},$$

where  $\Delta \mathbf{S}_{\text{event}}$  depends solely on the event type (death, birth, predation, reproduction, fight). Because all future consequences of events are explicitly scheduled in the FES (e.g., newborn death times, expected birth times of pregnant females), the system evolves in a fully event-driven and Markovian manner: the next state depends only on the current population configuration and the next event in the queue, without requiring any memory of past states.

### 4.4 Simulations and Confidence Intervals

Each test configuration is simulated  $N = 20$  independent times. In every run, small random perturbations (up to  $\pm 10\%$ ) are applied to reproduction, predation, and fight probabilities, introducing realistic stochastic variability.

For each metric (population sizes, births, deaths, predations, fights, competition events), the simulator computes the sample mean and the corresponding 95% confidence interval. Confidence intervals are obtained using the standard formula:

$$\text{CI}_{95} = \bar{x} \pm t_{0.95, N-1} \frac{\sigma}{\sqrt{N}},$$

where  $\bar{x}$  is the sample mean and  $t_{0.95, N-1}$  is the Student's  $t$  critical value.

For every test, the output consists of:

- a plot of the population trajectories (mean + variability),
- a table summarising the mean values and 95% confidence intervals of all metrics.

This approach enables statistically robust comparisons across different ecological scenarios.



## 5 Simulation Tests

We performed a series of simulation tests to explore the ecological dynamics between herbivore (H) and carnivore (C) populations under different parameter configurations. Each test varied one or more of the following key parameters:

- Initial fraction of herbivores ( $\text{perc\_herb}$ )
- Predation probability ( $p_{\text{pred}}$ )
- Reproduction probability ( $p_{\text{reprod}}$ )
- Intraspecific fight probability ( $p_{\text{fight}}$ )
- Herbivore competition threshold ( $\text{herb\_th}$ )
- Carnivore competition threshold ( $C_{\text{threshold}}$ )

Each simulation was repeated  $N = 20$  times to compute mean population trajectories, confidence intervals, and extinction frequencies. Here we summarize the parameter settings and expected behaviors for each ecological scenario.

### 5.1 Test 1: Baseline Predator–Prey Dynamics

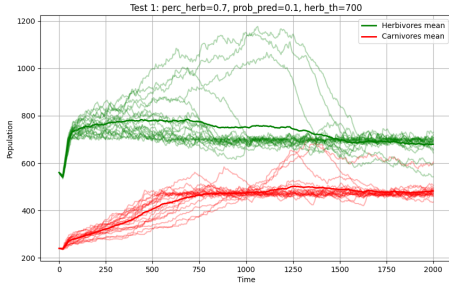


Figure 1: Population dynamics.  
**Parameters:**

- $N = 800$ ,  $\text{perc\_herb} = 0.7$
- $p_{\text{reprod}} = 0.7$ ,  $p_{\text{pred}} = 0.10$
- $p_{\text{fight}} = 0.1$
- $\text{herb\_th} = 700$ ,  $C_{\text{threshold}} = 1.5$

**Expected dynamics:** This scenario establishes a stable ecosystem where both populations coexist sustainably. A low predation probability ( $\text{prob}_{\text{pred}} = 0.10$ ) ensures that carnivores do not deplete the prey population too rapidly, while a minimal fight probability ( $\text{prob}_{\text{fight}} = 0.1$ ) reduces mortality from intraspecific aggression. This balance allows the dynamics to be driven primarily by resource availability, resulting in natural oscillations around the environment’s carrying capacity. This scenario serves as a reference for comparison with other parameter settings.

Test 1 Metrics Confidence Intervals (20 simulations)

| Metric             | Mean (95% CI)             |
|--------------------|---------------------------|
| meetings           | 37287.2 (36791.2-37783.1) |
| births             | 4709.9 (4616.2-4803.5)    |
| deaths_total       | 4339.9 (4255.9-4423.8)    |
| deaths_natural     | 288.4 (281.4-295.5)       |
| deaths_predation   | 1735.9 (1661.9-1809.9)    |
| deaths_fight       | 1001.0 (981.7-1020.2)     |
| deaths_competition | 1314.5 (1274.1-1355.0)    |
| fight              | 1001.0 (981.7-1020.2)     |
| predations         | 1735.9 (1661.9-1809.9)    |
| realized_natural_H | 178.8 (171.0-186.7)       |
| realized_natural_C | 109.6 (106.6-112.6)       |

Figure 2: Performance metrics.

## 5.2 Test 2: High Predation Pressure

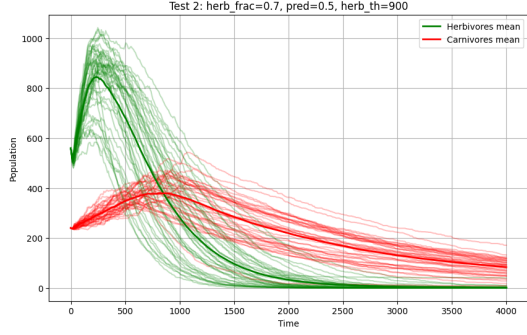


Figure 3: Population dynamics.

### Parameters:

- $N = 800$ ,  $\text{perc\_herb} = 0.7$
- $p_{\text{reprod}} = 0.6$ ,  $p_{\text{pred}} = 0.70$
- $p_{\text{fight}} = 0.1$
- $\text{herb\_th} = 800$ ,  $C_{\text{threshold}} = 1.1$

**Expected dynamics:** Driven by high predation efficiency ( $\text{prob}_p\text{red}=0.5$ ), the carnivore population surges and consumes the herbivores faster than they can reproduce, causing the prey to crash to extinction around  $t=2000$ . Following this collapse, the predators—having destroyed their own food source—enter a long, inevitable decline due to starvation, leading to total ecosystem failure.

Test 2 Metrics Confidence Intervals (40 simulations)

| Metric             | Mean (95% CI)          |
|--------------------|------------------------|
| meetings           | 7312.8 (6955.8-7669.9) |
| births             | 2245.4 (2037.1-2453.6) |
| deaths_total       | 3018.6 (2866.2-3171.0) |
| deaths_natural     | 1419.8 (1344.7-1495.0) |
| deaths_predation   | 1060.1 (989.1-1131.1)  |
| deaths_fight       | 231.0 (219.8-242.2)    |
| deaths_competition | 307.7 (288.9-326.5)    |
| fight              | 231.0 (219.8-242.2)    |
| predations         | 1060.1 (989.1-1131.1)  |
| realized_natural_H | 866.2 (820.8-911.5)    |
| realized_natural_C | 553.6 (522.9-584.4)    |

Figure 4: Performance metrics.

## 5.3 Test 3: Predator Dominance & Prey Decline

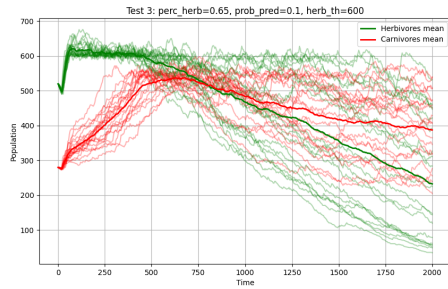


Figure 5: Population dynamics.

### Parameters:

- $N = 800$ ,  $\text{perc\_herb} = 0.65$
- $p_{\text{reprod}} = 0.7$ ,  $p_{\text{pred}} = 0.1$
- $p_{\text{fight}} = 0.2$
- $\text{herb\_th} = 600$ ,  $C_{\text{threshold}} = 1.1$

Test 3 Metrics Confidence Intervals (20 simulations)

| Metric             | Mean (95% CI)             |
|--------------------|---------------------------|
| meetings           | 29115.8 (27419.0-30812.7) |
| births             | 3993.8 (3730.6-4256.9)    |
| deaths_total       | 4174.1 (4005.6-4342.6)    |
| deaths_natural     | 224.1 (210.5-237.6)       |
| deaths_predation   | 1388.8 (1320.8-1456.9)    |
| deaths_fight       | 1494.9 (1422.6-1567.2)    |
| deaths_competition | 1066.3 (1026.3-1106.3)    |
| fight              | 1494.9 (1422.6-1567.2)    |
| predations         | 1388.8 (1320.8-1456.9)    |
| realized_natural_H | 114.8 (103.8-125.8)       |
| realized_natural_C | 109.2 (104.4-114.1)       |

Figure 6: Performance metrics.

**Expected dynamics:** Despite low predation efficiency (0.1), the permissive starvation threshold (1.1) allows the carnivore population to grow unchecked until it nearly equals the prey population. This leads to a "predator overshoot," where carnivores eventually outnumber herbivores, causing a subsequent decline in the prey population due to unsustainable pressure.

## 5.4 Test 4: Ecosystem Collapse (Hostile Environment)

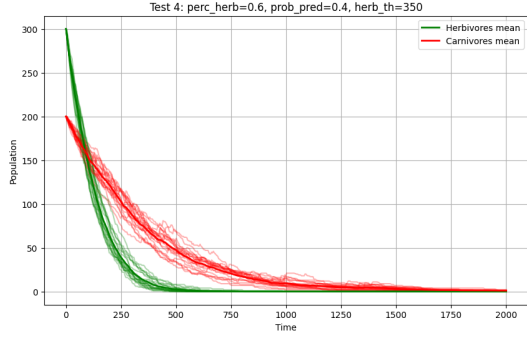


Figure 7: Population dynamics.

### Parameters:

- $N = 500$ ,  $\text{perc.herb} = 0.6$
- $p_{\text{reprod}} = 0.3$ ,  $p_{\text{birth}} = 0.5$
- $p_{\text{pred}} = 0.4$ ,  $p_{\text{fight}} = 0.4$
- $\text{scale.life} = 2000$
- $\text{herb.th} = 350$ ,  $C_{\text{threshold}} = 2.0$

**Expected dynamics:** This scenario simulates a hostile environment characterized by short lifespans, low fertility and high conflict rates. The combined pressures of natural mortality and predation prevent population recovery, leading to a rapid and joint extinction of both species.

Test 4 Metrics Confidence Intervals (20 simulations)

| Metric             | Mean (95% CI)          |
|--------------------|------------------------|
| meetings           | 1679.7 (1617.9-1741.5) |
| births             | 60.2 (50.6-69.9)       |
| deaths_total       | 559.6 (550.1-569.2)    |
| deaths_natural     | 13.1 (11.7-14.4)       |
| deaths_predation   | 254.8 (250.4-259.1)    |
| deaths_fight       | 201.2 (193.6-208.8)    |
| deaths_competition | 90.7 (87.8-93.5)       |
| fight              | 201.2 (193.6-208.8)    |
| predations         | 254.8 (250.4-259.1)    |
| realized_natural_H | 4.5 (3.9-5.1)          |
| realized_natural_C | 8.6 (7.4-9.7)          |

Figure 8: Performance metrics.

## 5.5 Test 5: Herbivore Dominance (Predator Extinction)

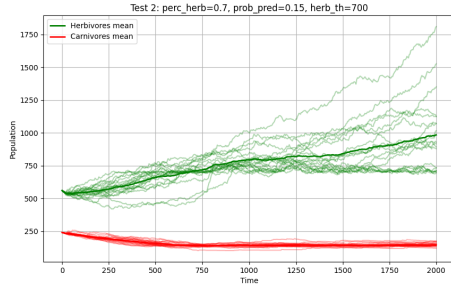


Figure 9: Population dynamics.

### Parameters:

- $N = 800$ ,  $\text{perc.herb} = 0.6$
- $p_{\text{reprod}} = 0.6$ ,  $p_{\text{birth}} = 0.85$
- $p_{\text{pred}} = 0.1$ ,  $p_{\text{fight}} = 0.2$
- $\text{herb.th} = 600$ ,  $C_{\text{threshold}} = 5.0$

Test 2 Metrics Confidence Intervals (20 simulations)

| Metric             | Mean (95% CI)             |
|--------------------|---------------------------|
| meetings           | 29368.4 (28232.2-30504.6) |
| births             | 3933.3 (3694.6-4172.0)    |
| deaths_total       | 3601.3 (3462.6-3740.1)    |
| deaths_natural     | 231.8 (220.3-243.3)       |
| deaths_predation   | 1231.5 (1167.2-1295.8)    |
| deaths_fight       | 1047.4 (989.3-1105.5)     |
| deaths_competition | 1090.7 (1019.6-1161.7)    |
| fight              | 1047.4 (989.3-1105.5)     |
| predations         | 1231.5 (1167.2-1295.8)    |
| realized_natural_H | 191.7 (180.8-202.6)       |
| realized_natural_C | 40.1 (37.5-42.7)          |

Figure 10: Performance metrics.

**Expected dynamics:** The high starvation threshold ( $C_{\text{threshold}} = 5$ ) places immediate pressure on the carnivore population, which starts below the sustainability ratio. This triggers a rapid extinction of predators. Consequently, the herbivores, released from predation pressure, become the dominant species, stabilizing around their resource threshold despite the short lifespans and low fertility rates.

## 5.6 Test 6: Herbivore-Dominated Initial State

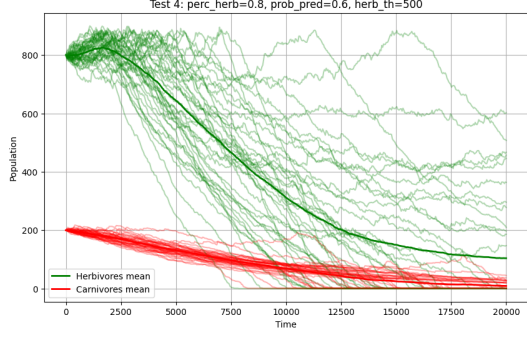


Figure 11: Population dynamics.

### Parameters:

- $N = 1000$ ,  $\text{perc\_herb} = 0.8$
- $p_{\text{reprod}} = 0.6$ ,  $p_{\text{pred}} = 0.3$
- $p_{\text{fight}} = 0.12$
- $\text{herb\_th} = 500$ ,  $C_{\text{threshold}} = 1.1$

**Expected dynamics:** The simulation begins with a high density of herbivores. This initial overabundance triggers two immediate effects. The high prey availability fuels a rapid growth in the carnivore population. The herbivore population (green line) immediately faces strong pressure from both resource competition and increasing predation. As prey becomes less super-abundant, the carnivore population also corrects downwards, eventually synchronizing with the prey population in a stable, lower-density equilibrium typical of a mature ecosystem.

Test 4 Metrics Confidence Intervals (40 simulations)

| Metric             | Mean (95% CI)          |
|--------------------|------------------------|
| meetings           | 5779.9 (5302.6-6257.2) |
| births             | 1563.0 (1358.3-1767.6) |
| deaths_total       | 2450.5 (2282.2-2618.8) |
| deaths_natural     | 1181.2 (1083.0-1279.5) |
| deaths_predation   | 976.9 (905.5-1048.2)   |
| deaths_fight       | 250.5 (225.2-275.8)    |
| deaths_competition | 41.9 (32.4-51.4)       |
| fight              | 250.5 (225.2-275.8)    |
| predations         | 976.9 (905.5-1048.2)   |
| realized_natural_H | 977.9 (886.6-1069.2)   |
| realized_natural_C | 203.3 (194.2-212.5)    |

Figure 12: Performance metrics.

## 5.7 Test 7: Strong Predation & Strong Carnivore Competition

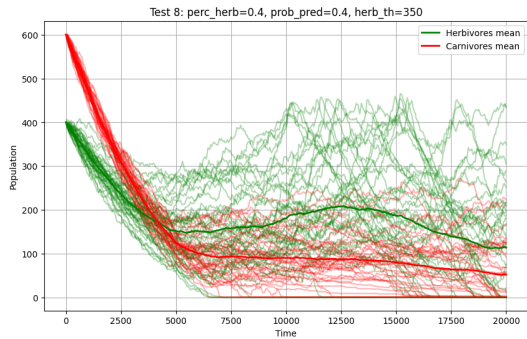


Figure 13: Population dynamics.

### Parameters:

- $N = 1000$ ,  $\text{perc\_herb} = 0.5$
- $p_{\text{reprod}} = 0.4$ ,  $p_{\text{pred}} = 0.55$
- $p_{\text{fight}} = 0.18$
- $\text{herb\_th} = 450$ ,  $C_{\text{threshold}} = 1.3$

**Expected dynamics:** In this scenario, the carnivore population faces strong resource competition immediately. Even if predation is efficient, the predators require a high surplus of prey

Test 8 Metrics Confidence Intervals (40 simulations)

| Metric             | Mean (95% CI)          |
|--------------------|------------------------|
| meetings           | 6360.9 (5900.7-6821.0) |
| births             | 2067.7 (1847.8-2287.6) |
| deaths_total       | 2901.5 (2711.5-3091.5) |
| deaths_natural     | 829.0 (763.2-894.7)    |
| deaths_predation   | 1112.6 (1020.8-1204.4) |
| deaths_fight       | 535.6 (497.4-573.8)    |
| deaths_competition | 424.4 (406.4-442.3)    |
| fight              | 535.6 (497.4-573.8)    |
| predations         | 1112.6 (1020.8-1204.4) |
| realized_natural_H | 468.6 (425.9-511.4)    |
| realized_natural_C | 360.3 (335.1-385.6)    |

Figure 14: Performance metrics.

to sustain reproduction. When this condition is not met, the carnivore population enters a phase of reproductive collapse and high starvation mortality. As the predator population crashes toward extinction, the predation pressure lifts, allowing the herbivore population—which may have been suppressed initially—to recover, grow rapidly, and eventually saturate the environment’s carrying capacity.

## 5.8 Test 8: Extreme Conditions Stress Test

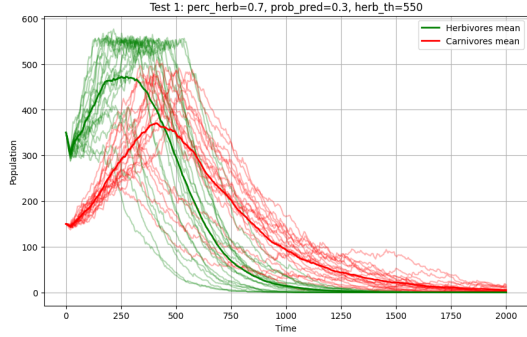


Figure 15: Population dynamics.

### Parameters:

- $N = 500$ ,  $\text{perc\_herb} = 0.7$
- $p_{\text{reprod}} = 0.9$ ,  $p_{\text{pred}} = 0.3$
- $p_{\text{fight}} = 0.75$
- $\text{herb\_th} = 550$ ,  $C_{\text{threshold}} = 1.1$

Test 1 Metrics Confidence Intervals (20 simulations)

| Metric             | Mean (95% CI)          |
|--------------------|------------------------|
| meetings           | 8500.3 (7583.0-9417.6) |
| births             | 2657.4 (2350.1-2964.7) |
| deaths_total       | 3151.8 (2845.9-3457.8) |
| deaths_natural     | 64.5 (57.3-71.8)       |
| deaths_predation   | 1029.9 (938.7-1121.1)  |
| deaths_fight       | 1808.2 (1619.3-1997.0) |
| deaths_competition | 249.2 (222.0-276.5)    |
| fight              | 1808.2 (1619.3-1997.0) |
| predations         | 1029.9 (938.7-1121.1)  |
| realized_natural_H | 30.9 (26.4-35.5)       |
| realized_natural_C | 33.6 (29.8-37.4)       |

Figure 16: Performance metrics.

**Expected dynamics:** This scenario applies severe stress through high reproduction rates and a high fight probability. Consequently, the general trend for both populations is negative. However, the decline is gradual rather than rapid. Despite the adverse parameters, the environment provides just enough sustenance to prevent an immediate collapse, allowing both species to linger and survive for an extended period before the slow depletion of resources takes its toll.

## 6 General Overview of Results

The summary table (Figure 17) provides a comprehensive overview of the system’s sensitivity to biological parameters.

Simulation Summary: Parameters vs Extinction Probability

| ID | Parameters                                                                                                                                                             | Extinct% |
|----|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|----------|
| 1  | name=test 1; num_animals=800; perc_herb=0.7; prob_reprod=0.7; prob_pred=0.1; prob_fight=0.1; herb_th=700; cam_th=1.5; total_time=2000                                  | 0%       |
| 2  | name=test 2; num_animals=800; perc_herb=0.7; prob_reprod=0.6; prob_pred=0.1; prob_fight=0.1; herb_th=800; cam_th=1.1; total_time=2000                                  | 100%     |
| 3  | name=test 3; num_animals=800; perc_herb=0.65; prob_reprod=0.7; prob_pred=0.1; prob_fight=0.2; herb_th=600; cam_th=1.1; total_time=2000                                 | 0%       |
| 4  | name=test 4; num_animals=500; perc_herb=0.6; prob_reprod=0.3; prob_birth=0.5; prob_pred=0.4; prob_fight=0.4; scale_life=2000; herb_th=350; cam_th=2.0; total_time=2000 | 100%     |
| 5  | name=test 5; num_animals=800; perc_herb=0.6; prob_reprod=0.6; prob_birth=0.85; prob_fight=0.2; prob_pred=0.1; herb_th=600; cam_th=5.0; total_time=2000                 | 0%       |
| 6  | name=test 6; num_animals=500; perc_herb=0.8; prob_reprod=0.6; prob_pred=0.3; prob_fight=0.12; herb_th=500; cam_th=1.1; total_time=2000                                 | 45%      |
| 7  | name=test 7; num_animals=500; perc_herb=0.6; prob_reprod=0.4; prob_pred=0.55; prob_fight=0.18; herb_th=450; cam_th=1.3; total_time=2000                                | 100%     |
| 8  | name=test 8; num_animals=500; perc_herb=0.7; prob_reprod=0.9; prob_pred=0.3; prob_fight=0.75; herb_th=550; cam_th=1.1; total_time=2000                                 | 100%     |

Figure 17: Aggregated performance metrics and extinction probabilities across all test scenarios.

The simulation data suggests that the ecosystem’s survival is not linear but depends on a delicate balance between parameters, thresholds and probabilities. By comparing the outcomes across different configurations, three fundamental ecological dynamics emerge:

- **Inefficient Predators work best:** We might think a better hunter is more successful, but the tests show the opposite. When predators are too lethal ( $p_{pred} > 0.5$ ), they eat all the prey too fast and then starve. To survive in the long run, predators must be "inefficient" ( $p_{pred} \approx 0.1$ ) so enough prey can escape and reproduce. It is a paradox.
- **High Food Requirements act as a Safety Brake:** The  $C_{threshold}$  is critical. If predators need a large surplus of food to reproduce (e.g., Test 5 with  $C_{threshold} = 5.0$ ), their population stops growing *before* they eat all the herbivores. This self-regulation prevents the ecosystem from crashing.
- **Reproduction is more important than Longevity:** In harsh environments (like Test 4), simply living a long time is not enough to survive. Populations need a high birth rate to "bounce back" from bad luck or sudden drops in numbers. Without new babies replacing the dead, the population slowly goes extinct.

## 7 Conclusions

The Markovian model developed in this work relies on the assumption that all events—natural deaths, predation, fights, reproduction and competition—occur with memoryless timings drawn from exponential distributions and that their outcomes depend solely on the current state of the system, including the counts of males, females, and pregnant females for both herbivores and carnivores. These assumptions justify treating the simulation as a discrete-event Markov process.

The resulting Markov chain is not strictly **ergodic**, as it contains multiple absorbing states: any state where  $H = 0$  or  $C = 0$  is absorbing for that species, since reproduction requires at least one male and one female and the fully extinct state with  $H = 0$  and  $C = 0$  is absorbing as no further births can occur. These absorbing states were directly observed in the simulation tests and extinction probabilities were quantified along with confidence intervals for population metrics across multiple runs.

The model could be generalized by relaxing the Markov assumptions, for instance by introducing memory effects where reproduction, predation or fights depend on past encounters, by allowing time-dependent rates reflecting seasonal or resource variations, or by incorporating spatial structure and history-dependent gestation or starvation effects. Such extensions would create a non-Markovian system capable of capturing more complex ecological dynamics, though at the cost of analytical simplicity. Nevertheless, the current Markovian framework is sufficient to investigate the qualitative behavior of predator-prey interactions and the effects of varying key parameters, as demonstrated by the simulation tests and the corresponding confidence interval analysis.